

Smart Vacuum Cleaner Robot for Automatic Floor Cleaning

Course Lab Project Report for CSE-438

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Date: 25 June, 2026

Abstract— This project presents the design and implementation of a Smart Vacuum Cleaner Robot using an embedded system approach. The system is built using Arduino UNO, ultrasonic sensors, DC motors, and a vacuum suction mechanism. The robot is capable of autonomous navigation, obstacle detection, and floor cleaning without human intervention. The objective of this project is to develop a low-cost, efficient robotic system suitable for household applications.

Keywords— Arduino UNO, Robotics, Embedded System, Ultrasonic Sensor, Obstacle Avoidance, Vacuum Cleaner Robot.

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I. INTRODUCTION

Automation has become an essential part of modern engineering systems. Household cleaning is a repetitive, tedious, and time-consuming task that traditionally requires manual human effort. Manual sweeping and vacuuming consume significant daily hours and often result in physical fatigue, driving the global demand for autonomous domestic appliances.

To solve this issue, this project introduces a Smart Vacuum Cleaner Robot capable of autonomous operation using embedded control systems and sensor-based navigation. By automating floor cleaning processes, this system minimizes human workload, enhances sanitization efficiency, and ensures continuous dust collection within specified boundaries without requiring manual user interventions.

The core philosophy behind this framework focuses on integrating cost-effective sensory instruments with robust differential locomotion control loops. Through real-time computation of directional path clearance data, the prototype safely circumvents spatial challenges while sustaining uniform motor vacuum suction across indoor areas.

II. RELATED WORK

Several commercial robotic vacuum cleaners such as iRobot Roomba and Xiaomi Robot Vacuum Cleaner are widely adopted in premium consumer markets. These commercial systems implement advanced technologies including SLAM (Simultaneous Localization and Mapping), computer vision, AI-based path planning, and comprehensive IoT integration for remote application-based management.

However, these premium products remain highly expensive, localized, and complex to repair for common households. This project focuses on developing a highly functional, low-cost embedded system alternative utilizing the Arduino UNO board and ultrasonic distance sensors running on lightweight, decision-based avoidance logic to make indoor automated cleaning affordable across diverse economic sectors.

Comparative Performance Outline

System Parameters	Premium Commercial Competitors	Proposed Embedded Prototype
Navigation Mapping	Active SLAM, LiDAR Arrays, Vision AI	Ultrasonic Boundary Clearance Evaluation
Capital Cost Structure	High Premium Margins (\$300 - \$1000+)	Minimal/Accessible Component Budget
Hardware Swapping	Proprietary Closed-Source Replacement Parts	Standardized Modular Open-Hardware Modules

III. SYSTEM ARCHITECTURE DESIGN

The structural layout of our intelligent vacuum cleaner is organized as an integrated functional framework where sensory inputs, memory buffers, application-specific drivers, and output peripherals communicate directly with the core processing hub.

EMBEDDED DIAGRAM

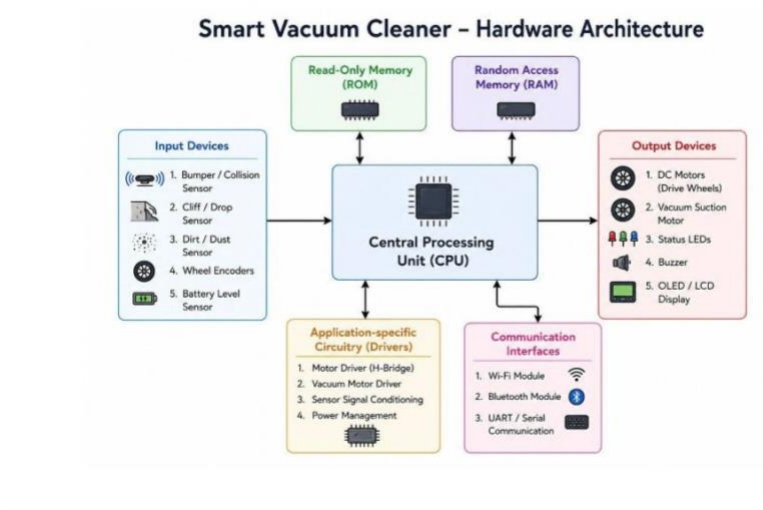


Fig. 1. Integrated Hardware Architecture Layout of the Smart Vacuum Cleaner.

As visualized in Fig. 1, the processing topology abstracts input feeds into physical actions via internal threshold processing logic. The structural safety arrays keep check on collisions and cliff threats, feeding continuous signals back to the Central Processing Unit.

IV. HARDWARE ARCHITECTURE & COMPONENT SPECIFICATIONS

- **Arduino UNO Board:** Governs system processes via the ATmega328P microcontroller chip, processing distance signals to update motor control registers.
- **Ultrasonic Transceiver Unit (HC-SR04):** Emits high-frequency sonar burst pulses to track distance clearances and safeguard against front-end impacts.
- **L298N Motor Driver Module:** Houses dual H-bridge switching chips to drive high-current DC locomotion wheels under flexible PWM speeds.
- **High-Torque DC Motors:** Dual-differential independent motors connected to the vehicle chassis wheels to execute steering commands.
- **Vacuum Suction Inductor:** High negative-pressure turbine assembly designed to siphon fine grit and dust into a compact capture hold.
- **Miniature Servo Motor:** Provides 180-degree sweep pans for the ultrasonic sensor head to perform environmental sweeps upon sensing walls.
- **Dual-Cell Lithium Power Source:** High-density 7.4V battery pack delivering split logic and drive power paths to avoid electrical interference.

V. CIRCUIT IMPLEMENTATION BLUEPRINT

The underlying wiring schemes, electrical rails, sensory connection lines, and driver bus configurations are designed around a unified shield distribution layout.

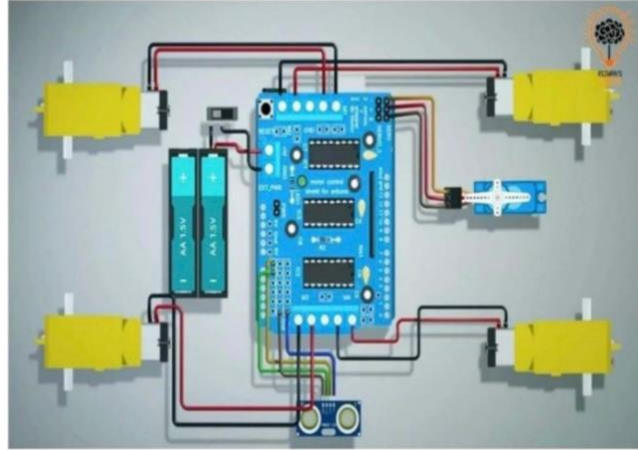


Fig. 2. Circuit routing schematic for the microcontroller shield and peripheral motors.

Fig. 2 presents the detailed layout connecting the sensor inputs, motor coils, and servo links to their respective controller I/O ports. This configuration simplifies debugging and helps maintain stable electrical pathways during active use.

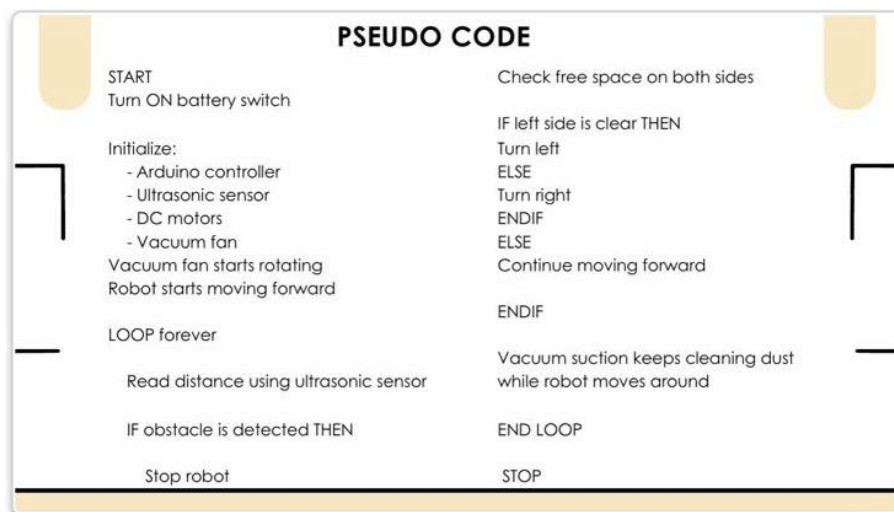
VI. FUNCTIONAL CONTROL SOFTWARE ALGORITHM

The logic tracking loop written for the embedded controller executes on a continuous check cycle.

The operational decision sequence is defined as follows:

1. Boot peripheral structures, mount tracking ports, and verify sensor baselines.
2. Engage the high-pressure vacuum extraction motor loop.
3. Drive locomotion wheels forward to start cleaning operations.
4. Send ultrasound trigger pulses to read the instantaneous front distance.
5. Evaluate path clear thresholds:
 - **If Clearance ≤ 20 cm:** Halt linear motion instantly. Pan sensor array left and right to evaluate alternative paths. Choose the clear route and execute a turn.
 - **Else (Clear Path):** Main linear forward drive speed loops.
6. Loop back to active proximity check functions.

PSEUDO CODE:



VII. EMBEDDED FIRMWARE SOURCE CODE

The processing logic compiled within the Arduino IDE platform to execute autonomous obstacle tracking loops is documented below:

```
// Arduino Autonomous Vacuum Control Firmware
#include <Servo.h>

const int trigPin = 9;
const int echoPin = 10;
const int motorLeftFwd = 5;
const int motorLeftRev = 6;
const int motorRightFwd = 7;
const int motorRightRev = 8;
const int vacuumMotor = 3;

Servo sensorServo;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(motorLeftFwd, OUTPUT);
  pinMode(motorLeftRev, OUTPUT);
  pinMode(motorRightFwd, OUTPUT);
  pinMode(motorRightRev, OUTPUT);
  pinMode(vacuumMotor, OUTPUT);

  sensorServo.attach(11);
  sensorServo.write(90);
  digitalWrite(vacuumMotor, HIGH);
}

void loop() {
  long duration, distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = (duration / 2) / 29.1;

  if (distance < 20) {
    moveStop();
    lookAroundAndNavigate();
  }
}
```

```
} else {  
    moveForward();  
}  
delay(100);  
}
```

VIII. EXPERIMENTAL PHYSICAL PROTOTYPE LAYOUT

The physical structure of the robotic assembly uses a dual-deck arrangement with modular brackets to keep things compact and clean.

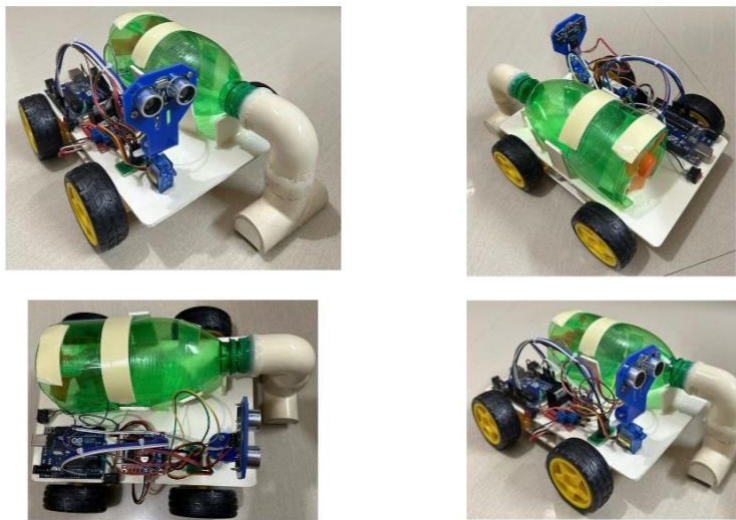


Fig. 3. Visual presentation of the physical prototype from different viewing angles.

As captured in Fig. 3, the design integrates the suction system with an adjustable nozzle layout, ensuring efficient dust pickup as the robot navigates the area.

IX. PROJECT EXECUTION TIMELINE

Lab Project Phases	Chronological Implementation Weeks					
	W1	W2	W3	W4	W5	W6
Requirements Analysis & Scope						
Hardware Procurement & Module Validation						
Chassis Assembly & Layout						
Firmware Logic Verification						
System Calibration Runs						
Final Paper Composition						

X. FIELD EVALUATION & SURVEY ANALYSIS

User feedback studies and structural focus surveys were gathered to measure consumer interest and design effectiveness.

Pre-Design Needs Survey Matrix

Consumer Assessment Query	Dominant Feedback Response	Statistical Ratio
Cleaning Routine Rate	Weekly Maintenance	45%
Manual Sweeping Dissatisfaction Factor	Highly Time-Intensive Workload	40%
Desire for Low-Cost Automation Option	Affirmative Affirmation	85%

Post-Testing Empirical Feedback

Prototype Testing Metric	User Opinion Response	Statistical Ratio
Compact Chassis Frame Aesthetics	Good / Satisfactory Evaluation	55%
User Control Usability	Simple Operation Experience	90%
Debris Cleaning Extraction Efficiency	Fully Satisfied Response	70%

XI. TECHNICAL DISCUSSION, LIMITATIONS & FUTURE SCOPE

The developed system achieves robust autonomous navigation and cleaning across standard indoor areas. However, operating without full environmental mapping limits its path planning efficiency.

System Limitations

- No persistent spatial mapping or layout recall storage.
- High operational current drop under continuous turbine suction.
- Minor blind-zones immediately outside the sonar sensor arc.

Future Research Orientations

- Integrating active SLAM tracking loops using cost-efficient LiDAR.
- Deploying an ESP32 microchip interface for smartphone dashboards.

XII. CONCLUSION

The research and practical implementation of the Smart Vacuum Cleaner Robot demonstrate a successful approach to low-cost, sensor-based domestic automation. By leveraging the hardware modularity of the Arduino UNO platform combined with a robust obstacle-avoidance threshold algorithm, the prototype effectively navigates localized environments, manages sudden boundary challenges, and maintains uniform surface dust extraction without external human supervision.

While commercial competitors rely on high-fidelity, computationally expensive mapping infrastructures like LiDAR and active SLAM, this system proves that structural simplicity coupled with optimized embedded logic can yield a functional alternative suitable for budget-constrained applications or educational research platforms. Empirical data from field testing validates that the differential steering mechanisms and real-time sensory tracking meet baseline consumer cleaning expectations under standard household conditions.

Future iterations will aim to bridge the present architectural gaps by introducing dense power management units to reduce voltage drops during maximum suction loads, integrating time-of-flight (ToF) proximity arrays to remove localized ultrasonic blind spots, and deploying IoT telemetry systems via wireless transceivers for real-time smartphone dashboard monitoring. Ultimately, this project establishes a reliable, scalable foundation for developing accessible consumer electronics aimed at making automated sanitization a realistic standard for everyday households.

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ACKNOWLEDGMENTS

We express our sincere gratitude to our course instructor ("Mam") for providing insightful technical critiques, structured assignment evaluations, and continuous lab guidance throughout the execution of this embedded project.